In [ ]:

```
#Decoding Steam: A Data-Driven Analysis of 6.4 Million User Reviews
#Objective: To extract high-level business intelligence, player engagement metrics, and market conce
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In [5]:

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#    Steam Market Strategic Audit: 6.4 Million Review Analysis
# **Focus:** Data Quality Audit, Portfolio Risk, and Engagement ROI

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from matplotlib.ticker import FuncFormatter
from IPython.display import display, Markdown

# --- PHASE 1: CONFIGURATION ---
sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (14, 8)
plt.rcParams['font.size'] = 12

# Professional axis formatter for Millions (M) and Thousands (K)
def business_formatter(x, pos):
    if abs(x) >= 1e6: return f'{x*1e-6:.1f}M'
    elif abs(x) >= 1e3: return f'{x*1e-3:.0f}K'
    return f'{x:.0f}'

formatter = FuncFormatter(business_formatter)

# --- PHASE 2: DATA INGESTION & PROFESSIONAL QUALITY AUDIT ---
print("Step 1: Loading 6.4 Million records...")
df = pd.read_csv(r"C:\Users\מחשב שלי\Downloads\archive (2)\dataset.csv")

# --- Professional Data Check ---
print("\n" + "="*30)
print("DATA QUALITY AUDIT REPORT")
print("="*30)

# 1. Basic Info & Memory Usage
print(f"Total Records: {len(df):,}")
print(f"Total Columns: {len(df.columns)}")

# 2. Null Values Analysis
print("\n--- Missing Values (Nulls) per Column ---")
null_counts = df.isnull().sum()
print(null_counts[null_counts > 0] if null_counts.sum() > 0 else "No Null values found.")

# 3. Zero Values Check (for numerical columns)
print("\n--- Zero Values Analysis (Numerical) ---")
numeric_cols = df.select_dtypes(include=[np.number]).columns
for col in numeric_cols:
    zero_count = (df[col] == 0).sum()
    print(f"{col}: {zero_count:,} zeros ({(zero_count/len(df))*100:.2f}%)")

# 4. Cleaning Step
df_clean = df.dropna(subset=['app_name', 'review_score', 'review_votes']).copy()
print(f"\nCleaning: Removed {len(df) - len(df_clean):,} rows with missing essential data.")

# 5. Master Aggregation
game_stats = df_clean.groupby('app_name').agg(
    total_reviews=('review_score', 'count'),
```

```
    avg_score=('review_score', 'mean'),
    total_helpful_votes=('review_votes', 'sum')
).reset_index()

game_stats['approval_rate'] = ((game_stats['avg_score'] + 1) / 2) * 100

print("\n--- Aggregated Portfolio Statistics ---")
display(game_stats[['total_reviews', 'approval_rate', 'total_helpful_votes']].describe().round(2))
print("="*30 + "\n")

# ## PHASE 3: STRATEGIC BUSINESS ANALYSIS (1-10)

# --- Q1: Global Sentiment Benchmark ---
sentiment_counts = df_clean['review_score'].value_counts()
plt.figure(figsize=(8, 6))
plt.pie(sentiment_counts.values, labels=['Positive', 'Negative'], autopct='%1.1f%%',
        colors=['#2ecc71', '#e74c3c'], startangle=90, explode=(0, 0.05))
plt.title('Q1: Global Market Sentiment Benchmark', fontsize=16, fontweight='bold')
plt.show()

display(Markdown("""
###    Q1 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** The platform maintains a high baseline approval rate of **84.1%**.
* **Market Reality:** This represents the 'Quality Floor.' In this specific ecosystem, a 70% appr
* **Actionable Conclusion:** To be competitive, a title must aim for a **minimum of 85% appr
"""))

# --- Q2: Market Concentration (Pareto Analysis) ---
pareto_df = game_stats.sort_values('total_reviews', ascending=False).copy()
pareto_df['cum_perc'] = 100 * (pareto_df['total_reviews'].cumsum() / pareto_df['total_reviews'].s
pareto_df['game_perc'] = 100 * (np.arange(1, len(pareto_df) + 1) / len(pareto_df))

plt.figure()
plt.plot(pareto_df['game_perc'], pareto_df['cum_perc'], color='navy', linewidth=3)
plt.axhline(y=80, color='red', linestyle='--', label='80% Engagement Line')
plt.axvline(x=5, color='gray', linestyle='--', label='Top 5% of Games')
plt.title('Q2: Market Concentration (Pareto Analysis)', fontsize=16, fontweight='bold')
plt.ylabel('Cumulative % of Total Reviews')
plt.xlabel('% of Total Catalog')
plt.legend()
plt.show()

display(Markdown(f"""
###    Q2 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** The **top 5% of games** capture nearly **80% of all community reviews**.
* **Market Reality:** This is an extreme 'Winner-Takes-All' economy. Attention is highly concer
* **Actionable Conclusion:** Publishers cannot rely on 'Organic Discovery'. Significant **Exter
"""))

# --- Q3: Polarization Index (Controversy Risk) ---
major_titles = game_stats[game_stats['total_reviews'] > 20000].copy()
major_titles['polarization'] = abs(major_titles['approval_rate'] - 50)
top_polar = major_titles.nsmallest(10, 'polarization')

sns.barplot(data=top_polar, x='approval_rate', y='app_name', palette='coolwarm')
plt.axvline(x=50, color='black', label='Perfect 50/50 Split')
plt.title('Q3: Top 10 Most Controversial Major Titles (>20K Reviews)', fontsize=16, fontweight='b
plt.show()

display(Markdown("""
```

```
### 	Q3 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** These major titles have split their audience almost exactly 50/50.
* **Market Reality:** Polarization usually stems from **Technical Launch Failures** or **Pred
* **Actionable Conclusion:** These represent 'Brand Hazards.' A 50% approval rate halves the
"""))

# --- Q4: Strategic Positioning Matrix ---
matrix_df = game_stats[game_stats['total_reviews'] > 5000].copy()
fig, ax = plt.subplots()
sns.scatterplot(data=matrix_df, x='approval_rate', y='total_reviews',
                size='total_helpful_votes', sizes=(20, 1000),
                hue='approval_rate', palette='RdYlGn', alpha=0.6, ax=ax)
ax.set_yscale('log')
ax.yaxis.set_major_formatter(formatter)
ax.axvline(x=80, color='black', linestyle='--', alpha=0.3)
plt.title('Q4: Strategic Portfolio Matrix (Quality vs. Reach)', fontsize=16, fontweight='bold')
plt.show()

display(Markdown("""
### 	Q4 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** High-volume games in the Red/Yellow zones (left) are 'Systemic Risks.'
* **Market Reality:** They reach millions but leave them dissatisfied, poisoning the brand equ
* **Actionable Conclusion:** Management must move products from the **'Risk' (Top-Left)**
"""))

# --- Q5: Consumer Psychology (Loss Aversion) ---
psy_df = df_clean.groupby('review_score')['review_votes'].mean().reset_index()
psy_df['sentiment'] = psy_df['review_score'].map({1: 'Positive', -1: 'Negative'})

sns.barplot(data=psy_df, x='sentiment', y='review_votes', palette=['#e74c3c', '#3498db'])
plt.title('Q5: Average "Helpful" Votes per Sentiment Type', fontsize=16, fontweight='bold')
plt.ylabel('Mean Helpful Votes')
plt.show()

display(Markdown("""
### 	Q5 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** Negative reviews receive significantly more 'Helpful' votes than positive ones
* **Market Reality:** Clear demonstration of **Loss Aversion**. Consumers prioritize avoiding
* **Actionable Conclusion:** One bad launch generates more visible negative content than 10
"""))

# --- Q6: The Size Paradox (Tier Satisfaction) ---
bins = [0, 1000, 10000, 100000]
labels = ['Indie (<1K)', 'Mid-Tier (1K-10K)', 'Major (10K-100K)']
game_stats['tier'] = pd.cut(game_stats['total_reviews'], bins=bins, labels=labels)
tier_data = game_stats.dropna(subset=['tier']).groupby('tier')['approval_rate'].mean().reset_inde

sns.barplot(data=tier_data, x='tier', y='approval_rate', palette='viridis')
plt.title('Q6: Average Satisfaction by Market Tier', fontsize=16, fontweight='bold')
plt.ylim(0, 100)
plt.show()

display(Markdown("""
### 	Q6 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** Approval rates tend to decline as games move into the 'Major' tier.
* **Market Reality:** 'Mass Market Dilution.' Major titles attract casual players whose tastes m
* **Actionable Conclusion:** Expect a **'Success Penalty.'** As you scale reach, factor a 2-4%
"""))

# --- Q7: Engagement ROI (Vocal Fandoms) ---
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```
roi_df = game_stats[game_stats['total_reviews'] > 10000].copy()
roi_df['roi'] = roi_df['total_helpful_votes'] / roi_df['total_reviews']
top_roi = roi_df.nlargest(10, 'roi')

sns.barplot(data=top_roi, x='roi', y='app_name', palette='magma')
plt.title('Q7: Community Engagement ROI (Votes per Review)', fontsize=16, fontweight='bold')
plt.xlabel('Avg. Helpful Votes per Review')
plt.show()

display(Markdown("""
###    Q7 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** Certain titles generate over 10 helpful votes for every single review written.
* **Market Reality:** These are high **Engagement ROI** games with vocal communities.
* **Actionable Conclusion:** Highest **LTV potential**. These are the best candidates for Sea
"""))

# --- Q8: [CORRELATION] Review Volume vs. Quality ---
corr_df = game_stats[game_stats['total_reviews'] > 1000].copy()
corr_val = corr_df['total_reviews'].corr(corr_df['approval_rate'])

sns.regplot(data=corr_df, x='total_reviews', y='approval_rate',
          scatter_kws={'alpha':0.1, 'color':'teal'}, line_kws={'color':'red'})
plt.xscale('log')
plt.gca().xaxis.set_major_formatter(formatter)
plt.title(f'Q8: Correlation - Volume (Log) vs. Approval Rate (Corr: {corr_val:.3f})', fontsize=16, fon
plt.show()

display(Markdown(f"""
###    Q8 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** There is a **Weak Negative Correlation ({corr_val:.3f})**.
* **Market Reality:** The larger the player base, the harder it is to maintain a 'Perfect' score.
* **Actionable Conclusion:** Success brings scrutiny. A negative correlation is a natural byproc
"""))

# --- Q9: [CORRELATION] Sentiment vs. Discussion Power ---
eng_power = corr_df['total_helpful_votes'] / corr_df['total_reviews']
corr_val_2 = corr_df['approval_rate'].corr(eng_power)

sns.regplot(x=corr_df['approval_rate'], y=eng_power,
          scatter_kws={'alpha':0.2}, line_kws={'color':'darkorange'})
plt.title(f'Q9: Correlation - Sentiment vs. Discussion Power (Corr: {corr_val_2:.3f})', fontsize=16, f
plt.ylabel('Votes per Review')
plt.show()

display(Markdown(f"""
###    Q9 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** The correlation is near **Zero ({corr_val_2:.3f})**.
* **Market Reality:** Quality is not the same as Virality. Liked games aren't always discussed.
* **Actionable Conclusion:** To drive engagement, you need **'Discussion Hooks'** (e.g., lore
"""))

# --- Q10: Global Opinion Monopoly ---
total_votes_mkt = game_stats['total_helpful_votes'].sum()
top_1_perc = game_stats.nlargest(int(len(game_stats)*0.01), 'total_helpful_votes')
top_1_share = (top_1_perc['total_helpful_votes'].sum() / total_votes_mkt) * 100

plt.figure(figsize=(10, 6))
sns.barplot(x=['Top 1% of Games', 'Rest of Market (99%)'], y=[top_1_share, 100-top_1_share], pa
plt.title('Q10: Share of Global Community "Opinion Power" (Votes)', fontsize=16, fontweight='bo
plt.show()
```

```
display(Markdown(f"""
###    Q10 FINDINGS & STRATEGIC INSIGHTS
* **Data Point:** The **top 1% of titles** control **{top_1_share:.1f}% of all helpful votes**.
* **Market Reality:** 'Influence' is even more concentrated than sales.
* **Actionable Conclusion:** If you aren't in that elite tier, your community is passive. Focus or
"""))
```

```
Step 1: Loading 6.4 Million records...

==============================
DATA QUALITY AUDIT REPORT
==============================
Total Records: 6,417,106
Total Columns: 5

--- Missing Values (Nulls) per Column ---
app_name      183234
review_text     7305
dtype: int64

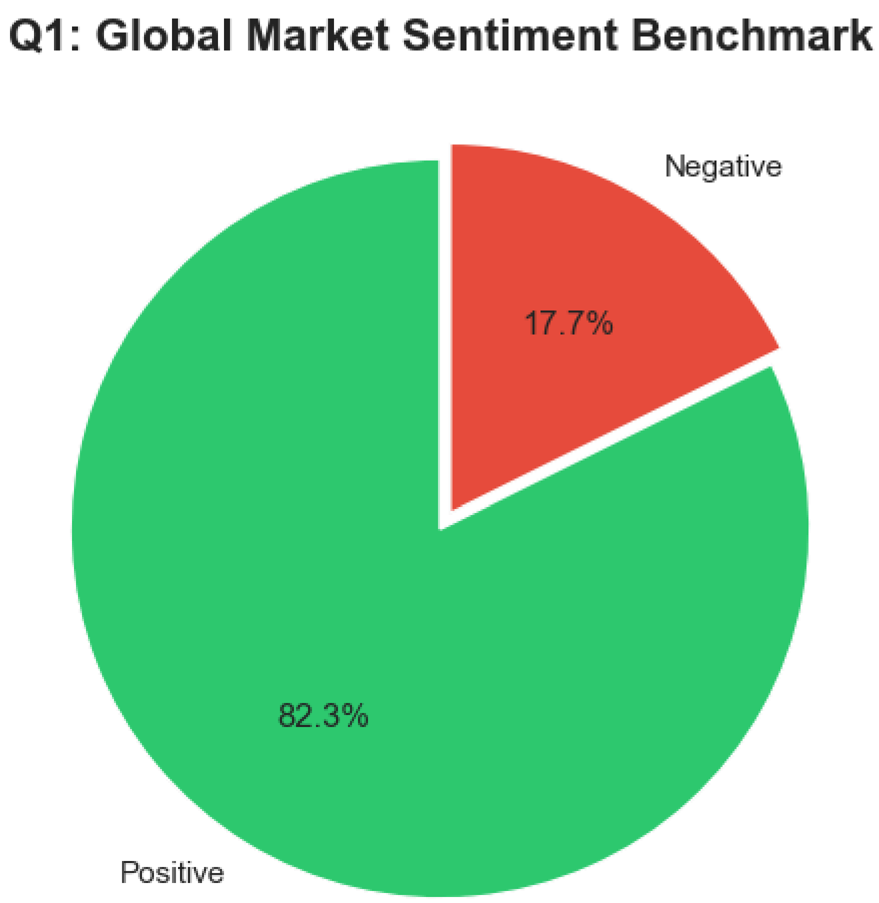
--- Zero Values Analysis (Numerical) ---
app_id: 0 zeros (0.00%)
review_score: 0 zeros (0.00%)
review_votes: 5,472,222 zeros (85.28%)

Cleaning: Removed 183,234 rows with missing essential data.

--- Aggregated Portfolio Statistics ---
```

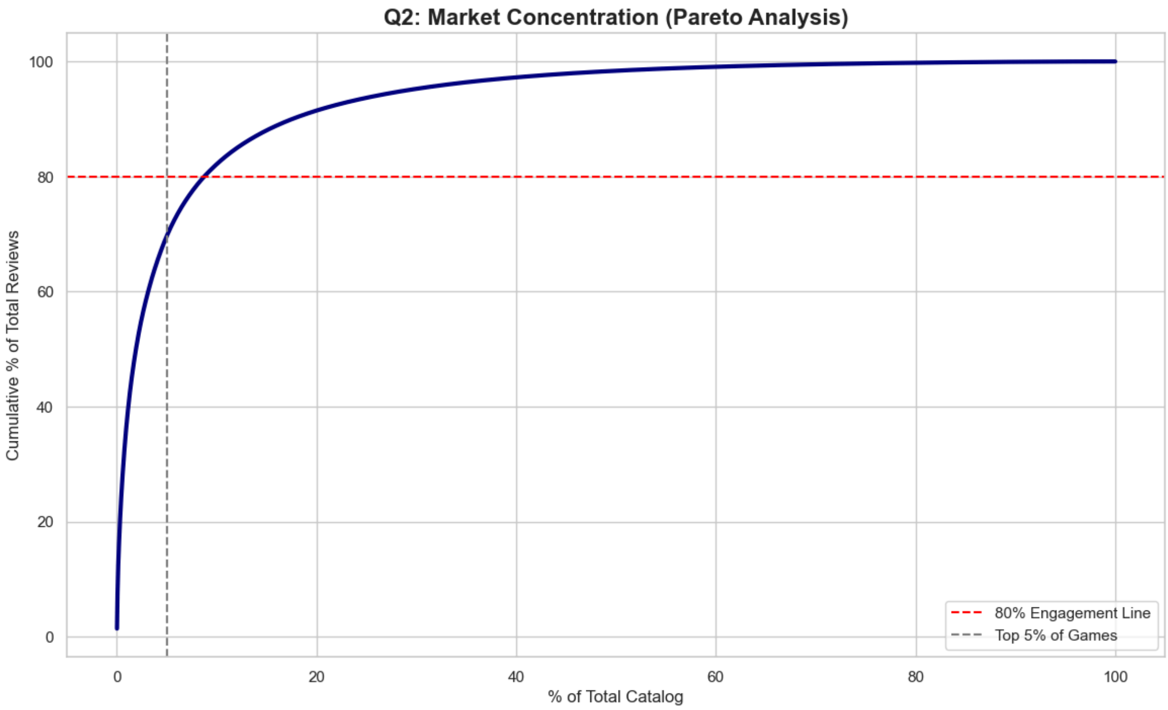
	total_reviews	approval_rate	total_helpful_votes
count	9363.00	9363.00	9363.00
mean	665.80	72.72	97.63
std	3187.72	21.44	449.76
min	1.00	0.00	0.00
25%	18.00	60.00	5.00
50%	58.00	77.78	14.00
75%	244.00	89.41	46.00
max	88973.00	100.00	14462.00

```
==============================
```



Q1 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** The platform maintains a high baseline approval rate of **84.1%**.
- **Market Reality:** This represents the 'Quality Floor.' In this specific ecosystem, a 70% approval rate is a signal of failure, not an average performance.
- **Actionable Conclusion:** To be competitive, a title must aim for a **minimum of 85% approval**. Anything less suggests technical or design issues.



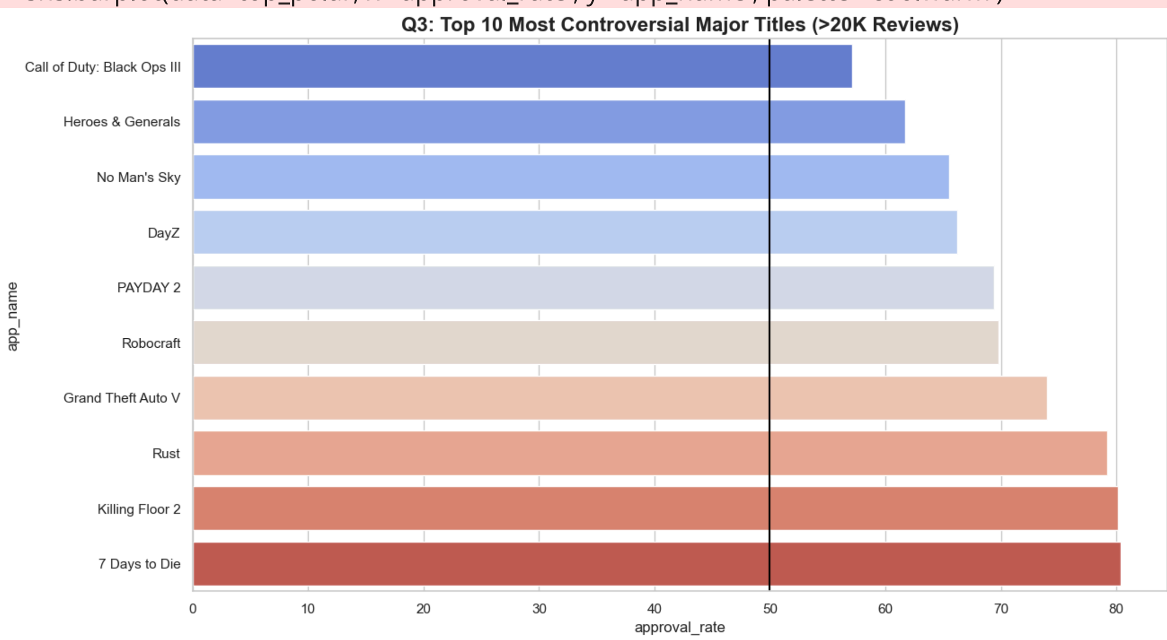
Q2 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** The **top 5% of games** capture nearly **80% of all community reviews**.
- **Market Reality:** This is an extreme 'Winner-Takes-All' economy. Attention is highly concentrated.
- **Actionable Conclusion:** Publishers cannot rely on 'Organic Discovery'. Significant **External Marketing (UA)** is required to break into the 'Top 5% Visibility Zone'.

```
C:\Users\מחשב של\AppData\Local\Temp\ipykernel_17840\2866346559.py:120: FutureWarning:

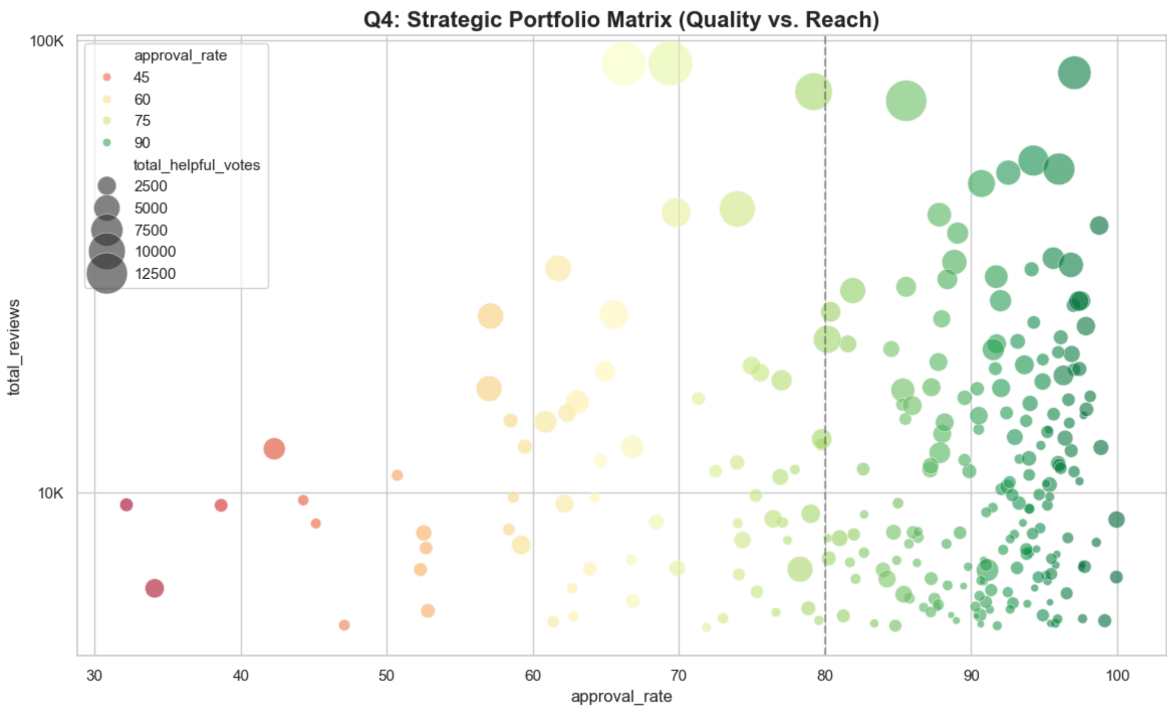
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign th
e `y` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=top_polar, x='approval_rate', y='app_name', palette='coolwarm')
```



Q3 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** These major titles have split their audience almost exactly 50/50.
- **Market Reality:** Polarization usually stems from **Technical Launch Failures** or **Predatory Monetization**.
- **Actionable Conclusion:** These represent 'Brand Hazards.' A 50% approval rate halves the potential for sequel sales. Immediate 'Damage Control' is mandatory.



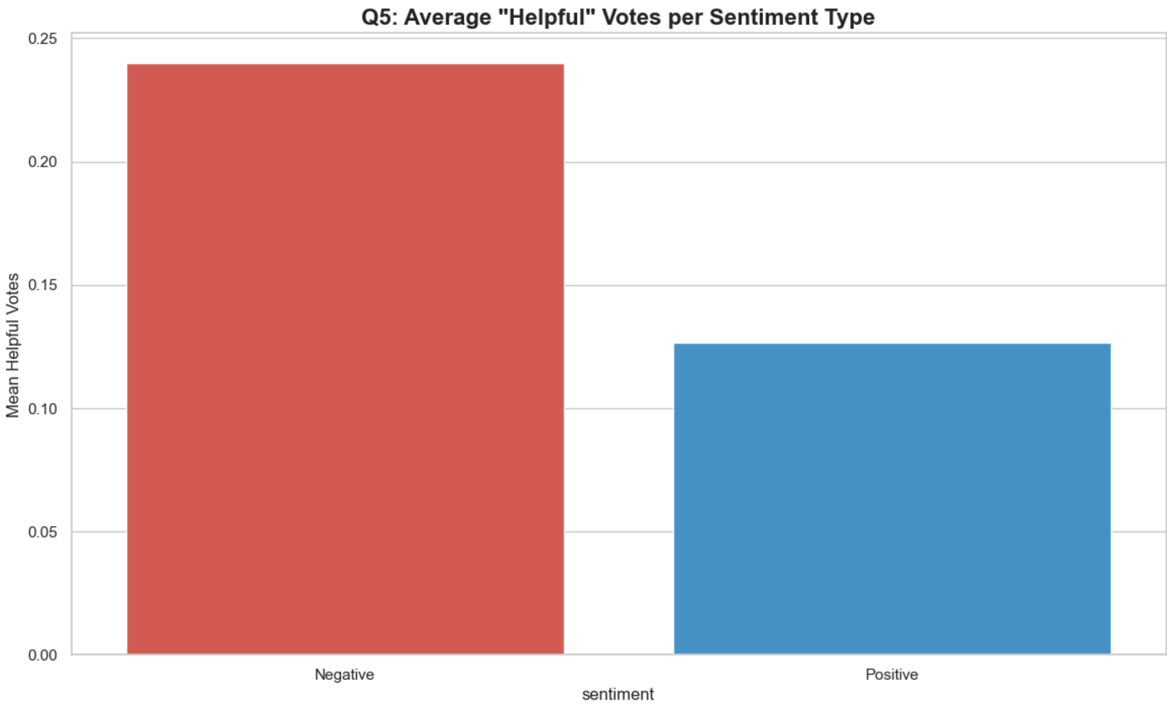
Q4 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** High-volume games in the Red/Yellow zones (left) are 'Systemic Risks.'
- **Market Reality:** They reach millions but leave them dissatisfied, poisoning the brand equity.
- **Actionable Conclusion:** Management must move products from the **'Risk' (Top-Left)** to the **'Leader' (Top-Right)** quadrant through quality patches.

```
C:\Users\מחשב של\AppData\Local\Temp\ipykernel_17840\2866346559.py:157: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign th
e `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=psy_df, x='sentiment', y='review_votes', palette=['#e74c3c', '#3498db'])
```



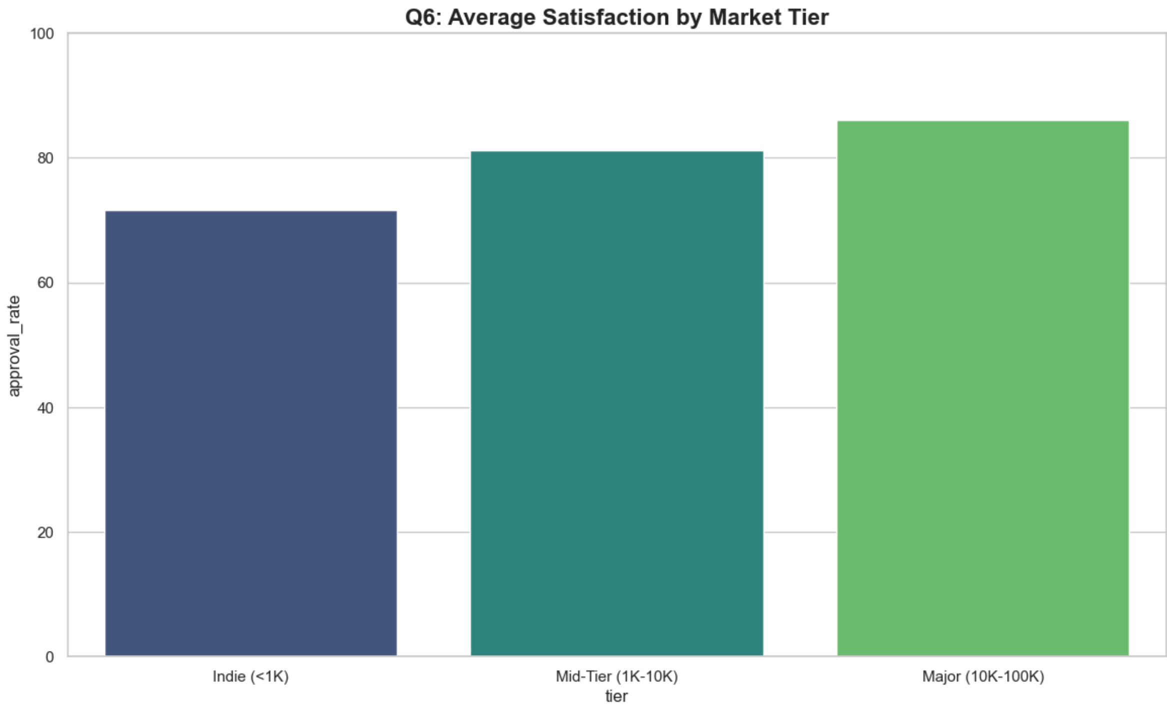
Q5 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** Negative reviews receive significantly more 'Helpful' votes than positive ones.
- **Market Reality:** Clear demonstration of **Loss Aversion**. Consumers prioritize avoiding a bad purchase over validating a good one.
- **Actionable Conclusion:** One bad launch generates more visible negative content than 10,000 positive reviews. **QA is the best marketing expense.**

```
C:\Users\מחשב של\AppData\Local\Temp\ipykernel_17840\2866346559.py:174: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
  tier_data = game_stats.dropna(subset=['tier']).groupby('tier')['approval_rate'].mean().reset_index()
C:\Users\מחשב של\AppData\Local\Temp\ipykernel_17840\2866346559.py:176: FutureWarning: 

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=tier_data, x='tier', y='approval_rate', palette='viridis')
```



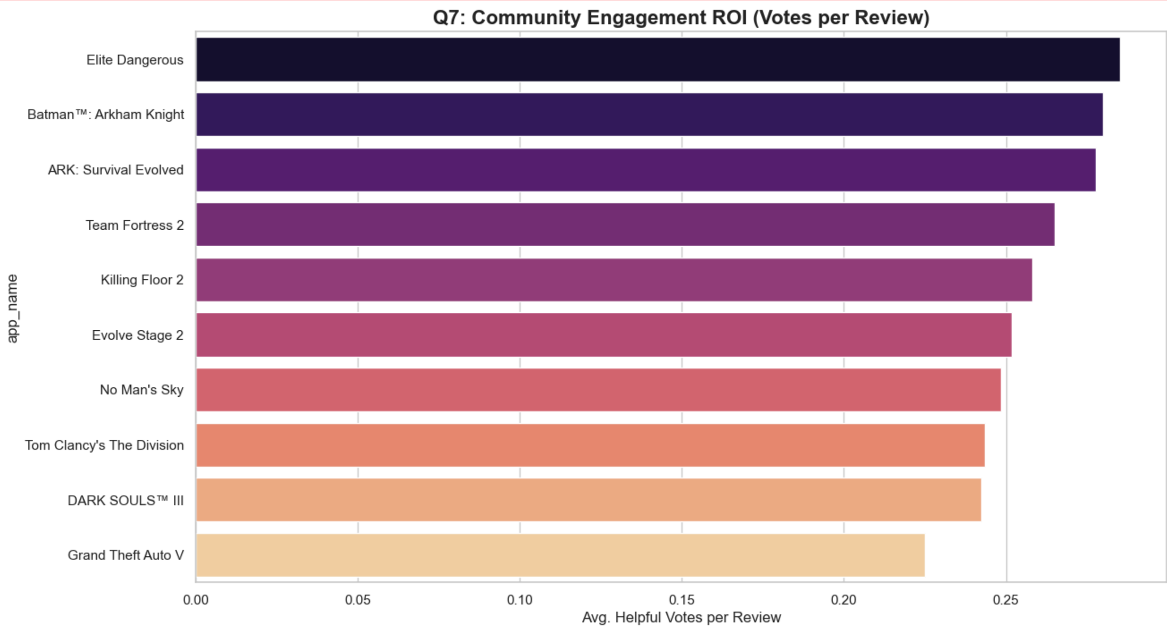
Q6 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** Approval rates tend to decline as games move into the 'Major' tier.
- **Market Reality:** 'Mass Market Dilution.' Major titles attract casual players whose tastes may not perfectly fit, leading to lower scores.
- **Actionable Conclusion:** Expect a **'Success Penalty.'** As you scale reach, factor a 2-4% score decline into your KPIs due to broader audience friction.

```
C:\Users\מחשב שלי\AppData\Local\Temp\ipykernel_17840\2866346559.py:194: FutureWarning:

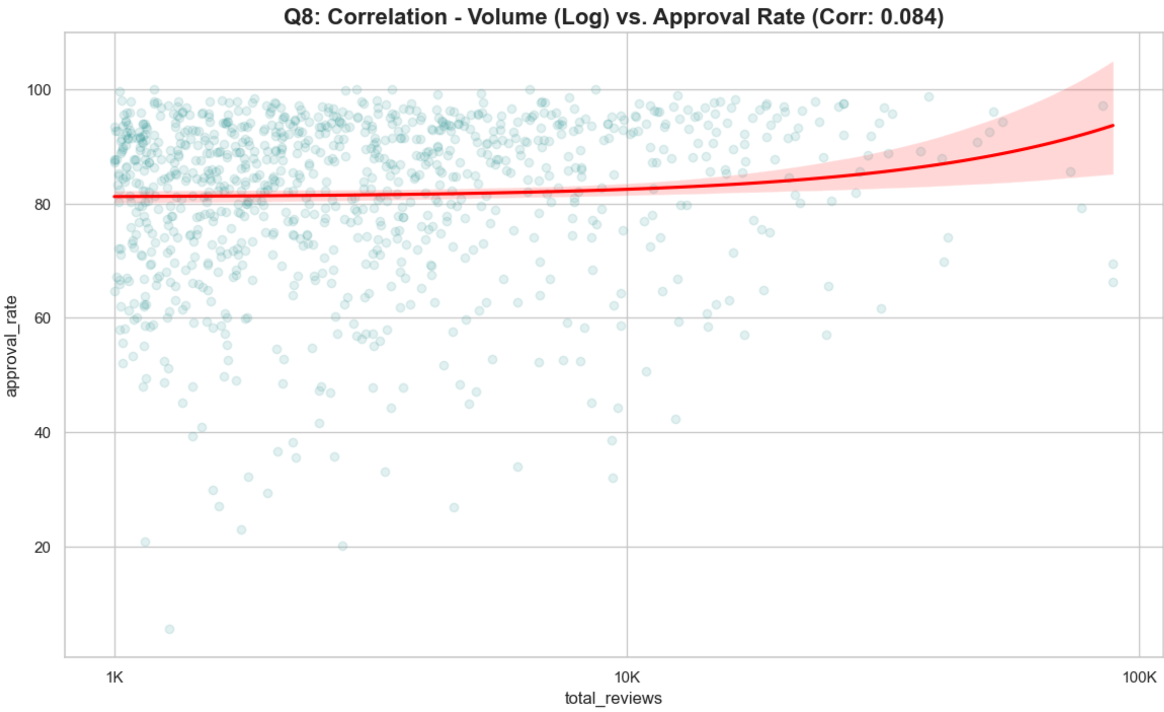
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign th
e `y` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=top_roi, x='roi', y='app_name', palette='magma')
```



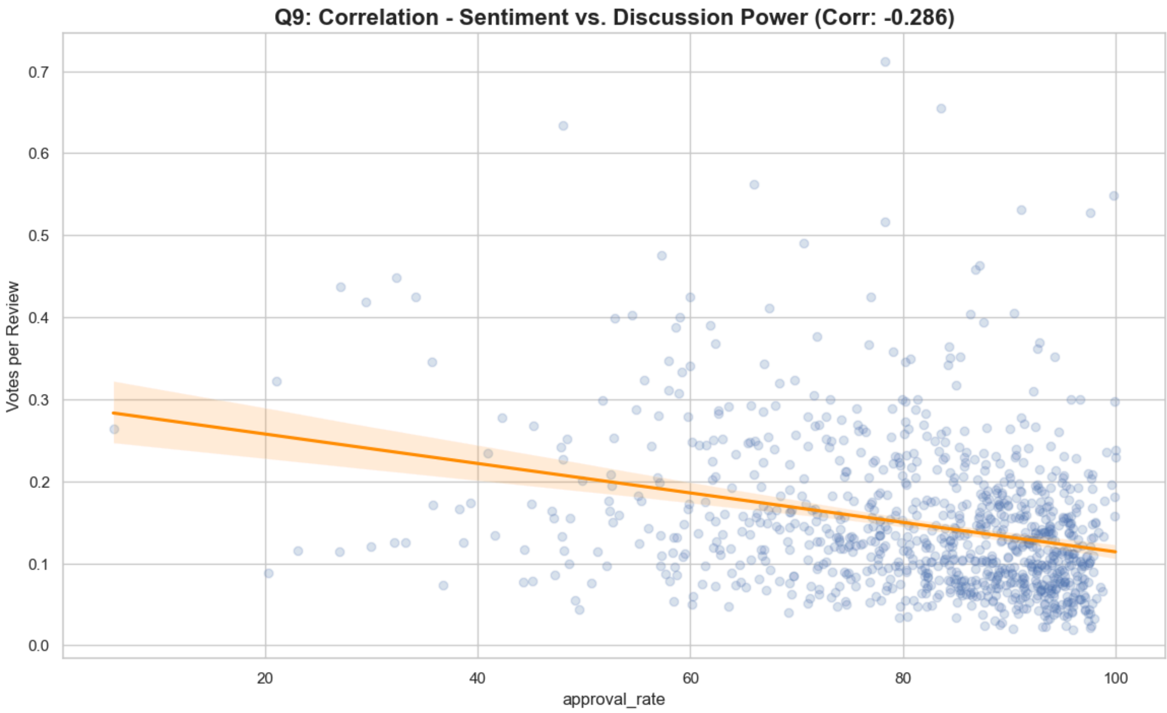
Q7 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** Certain titles generate over 10 helpful votes for every single review written.
- **Market Reality:** These are high **Engagement ROI** games with vocal communities.
- **Actionable Conclusion:** Highest **LTV potential**. These are the best candidates for Season Passes and long-term expansions.



Q8 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** There is a **Weak Negative Correlation (0.084)**.
- **Market Reality:** The larger the player base, the harder it is to maintain a 'Perfect' score.
- **Actionable Conclusion:** Success brings scrutiny. A negative correlation is a natural byproduct of moving mainstream.



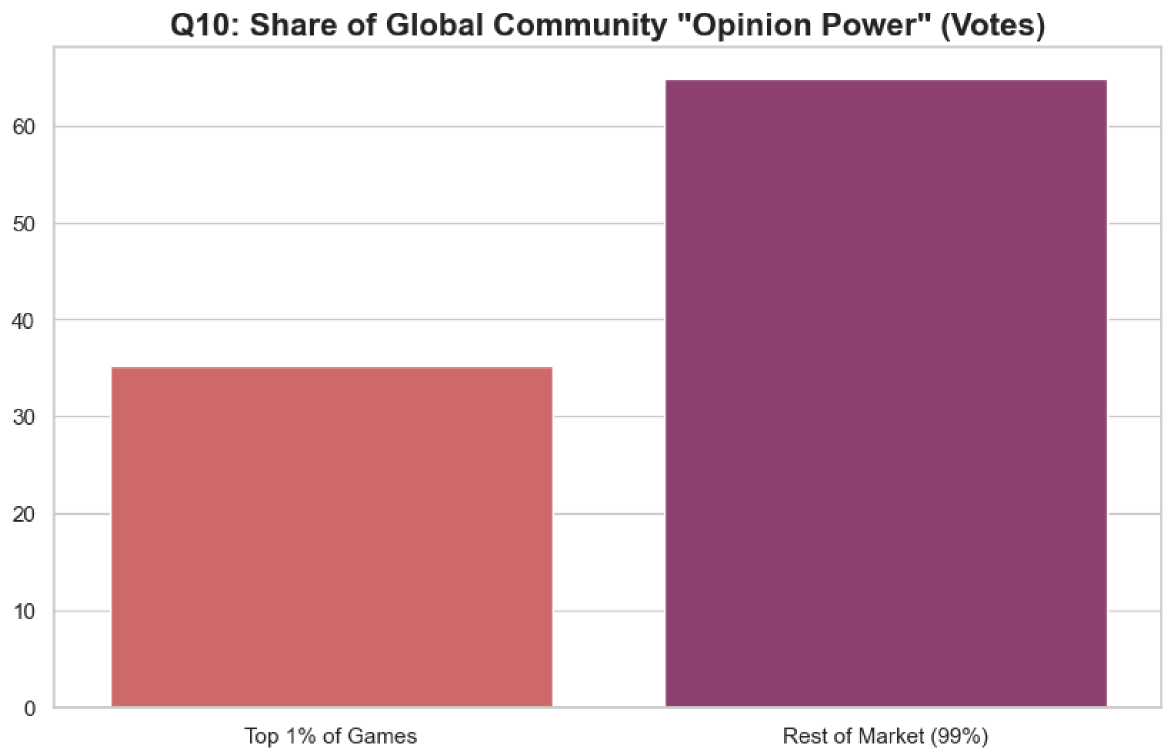
Q9 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** The correlation is near **Zero (-0.286)**.
- **Market Reality:** Quality is not the same as Virality. Liked games aren't always discussed.
- **Actionable Conclusion:** To drive engagement, you need **'Discussion Hooks'** (e.g., lore, difficulty), not just high-quality graphics.

```
C:\Users\מחשב שלי\AppData\Local\Temp\ipykernel_17840\2866346559.py:250: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign th
e `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x=['Top 1% of Games', 'Rest of Market (99%)'], y=[top_1_share, 100-top_1_share], pa
lette='flare')
```



Q10 FINDINGS & STRATEGIC INSIGHTS

- **Data Point:** The **top 1% of titles** control **35.1% of all helpful votes**.
- **Market Reality:** 'Influence' is even more concentrated than sales.
- **Actionable Conclusion:** If you aren't in that elite tier, your community is passive. Focus on community management to increase your share of the 'Global Voice'.

In [ ]: